

Formation MATHSTIC-BO

Part I – Module LP/ILP

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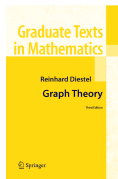
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May 2026

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- 4 Transition to practice

References



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As calculus developed from the seventeenth century's need to solve problems of mechanics, linear programming developed from the twentieth century's need to solve problems of management.

Vašek Chvátal

Addressing a decision-making problem is a two-step process:

① **Modeling**

translating the problem statement into a mathematical model.

② **Solving**

applying linear programming techniques to solve the problem.

Linear Programming refers to a set of powerful techniques for solving optimization problems that relies on a “linear” model of the addressed problem.

Linear Programming **is definitely not** application-dependent. It is successfully used to address a wide variety of problems in:

- Economy
- Agriculture
- Finance
- Public Services
- Transportation
- Production
- Logistics
- Planning and Scheduling
- Design
- And many other fields

A particular attention should be paid to modeling: if the model does not fit reality, even the most advanced resolution techniques cannot yield relevant solutions.

Modeling a problem is:

- ➊ Choosing and defining **decisions variables**
- ➋ Identifying the problem **constraints**, and stating these constraints as equalities or inequalities
- ➌ Identifying the problem **objective**, and stating this objective as a function to maximize (or to minimize)

Modeling example

A firm produces two product types A and B. Three raw materials (referred to as RM_1 , RM_2 and RM_3) are involved in the production process but are available in limited quantities every day.

	RM_1	RM_2	RM_3
A	2	1	0
B	1	2	1
Avail.	8	7	3

Producing and selling a single unit of product of type A yields 4 €, and 5 € for B.

Which products should be processed to maximize the total profit?

The production process is supposed to be linear.

Modeling example (con't)

Decision variables:

Constraints:

Objective:

Modeling example (con't)

Decision variables:

x_A is the quantity of product of type A to produce

x_B is the quantity of product of type B to produce

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Constraints: Raw materials are available in limited quantities

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$$2x_A + x_B \leq 8 \quad RM_1 \text{ availability}$$

$$x_A + 2x_B \leq 7 \quad RM_2 \text{ availability}$$

$$x_B \leq 3 \quad RM_3 \text{ availability}$$

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Decision variables are nonnegative

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Decision variables are nonnegative

$$x_A \geq 0 \quad x_A \text{ is nonnegative}$$

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Objective:

Maximize the profit

$$\text{Maximize } Z = 4x_A + 5x_B$$

The objective function and all the constraints are gathered to form a **linear program**:

$$(P) = \left\{ \begin{array}{l} \text{Maximize } Z = 4x_A + 5x_B \\ \text{subject to (s.t)} \\ 2x_A + x_B \leq 8 \\ x_A + 2x_B \leq 7 \\ x_B \leq 3 \\ x_A \geq 0 \\ x_B \geq 0 \end{array} \right.$$

A linear program is said to be **linear** because:

- constraints are linear

Each constraint can be written as $\sum_{j=1}^n a_j x_j \leq \text{or } = \text{or } \geq b$

Where x_j is a decision variable and a_j is a constant for all $j \in \{1, \dots, n\}$.

Note:

- The objective function is also linear

It can be written as Maximize (or Minimize) $\sum_{j=1}^n c_j x_j$

Where c_j is a constant for all $j \in \{1, \dots, n\}$.

Note:

Furthermore, decision variables have to be real; if they are boolean or integer, we have an Integer Linear Program.

Linear programming

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Note: $x_1 x_2 \leq 5$ is not a linear constraint

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Note: Maximize $x_1^2 - 5x_2$ is not a linear objective function

Furthermore, decision variables have to be real; if they are boolean or integer, we have an Integer Linear Program.

Statuses of a linear program

Once in its standard form, a linear program has exactly one of the three following statuses:

- **Unfeasible**

There is no solution because some constraints are conflicting.

Example:

- **Unbounded**

There are solutions but there is **no optimal** solution.

Example:

- **Feasible**

There exists an **optimal solution** i.e. a solution with bounded values for decision variables such that the objective function value is maximized (or minimized).

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There are solutions but there is **no optimal** solution.

Example:
$$\begin{cases} \text{Maximize } x_1 \\ x_1 - x_2 \geq 4 \end{cases} \quad (\text{with } x_1, x_2 \geq 0)$$

- **Feasible**

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An unfeasible or unbounded linear program is either the result of a bad modeling (most typical case) or a ill-defined problem.

Graphical method

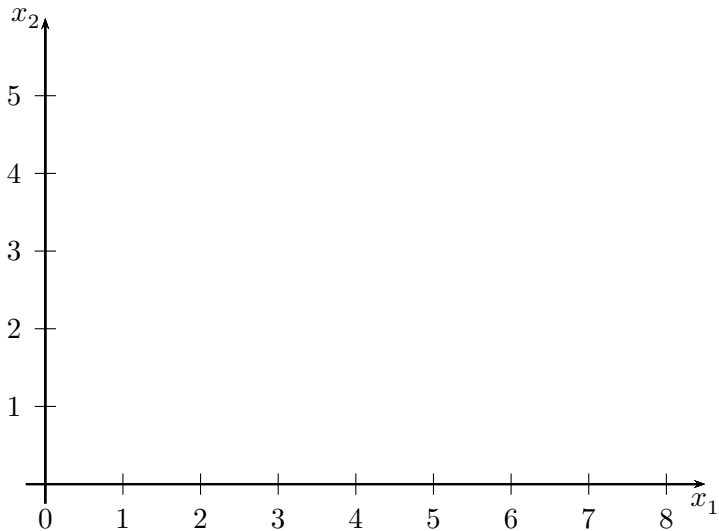
When there are only **two decision variables** in a linear program, it can be solved graphically by plotting the constraints in the plane and finding an optimal basis by moving a line modeling the objective function.

The set of feasible solutions is **convex**: if X and Y are feasible solutions to a LP, then so is $\alpha X + (1 - \alpha)Y$ for all $\alpha \in [0, 1]$. In other words, any point in the segment $[XY]$ is a feasible solution.

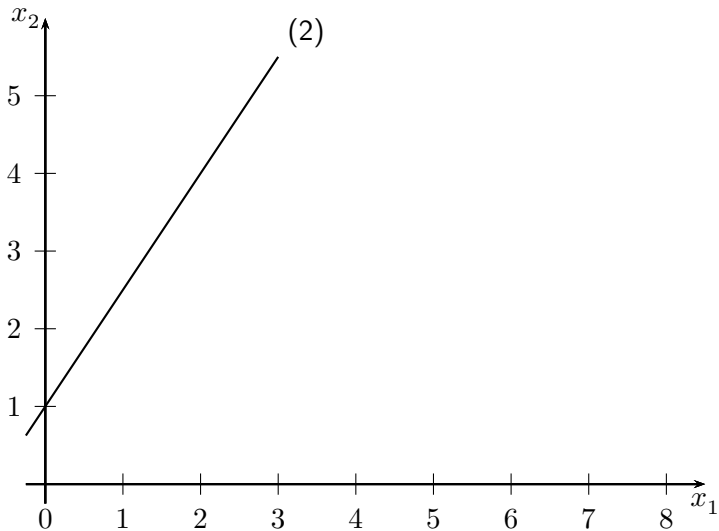
$$\left\{ \begin{array}{ll} \text{Maximize } Z = x_1 + 2x_2 & (1) \\ -3x_1 + 2x_2 \leq 2 & (2) \\ -x_1 + 2x_2 \leq 4 & (3) \\ x_1 + x_2 \leq 5 & (4) \\ x_1 \leq 3 & (5) \\ x_1, x_2 \geq 0 & (6) \end{array} \right.$$

The constraints' frontier are plotted (i.e. inequalities are replaced by equalities) for determining S , the set of all feasible solutions.

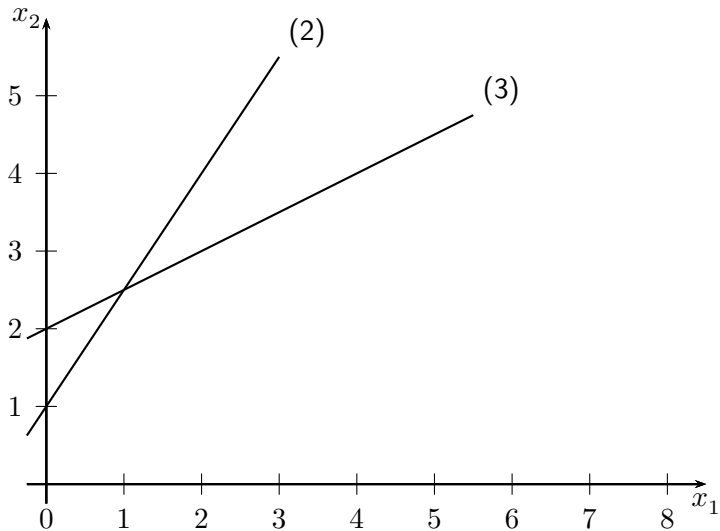
Graphical method



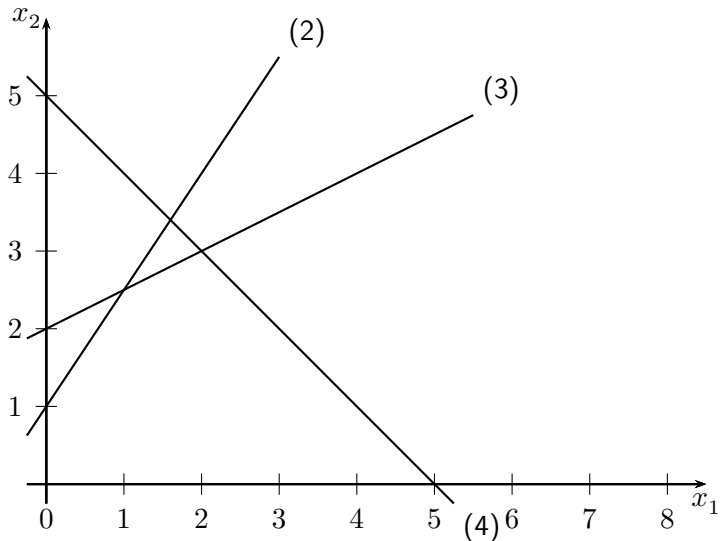
Graphical method



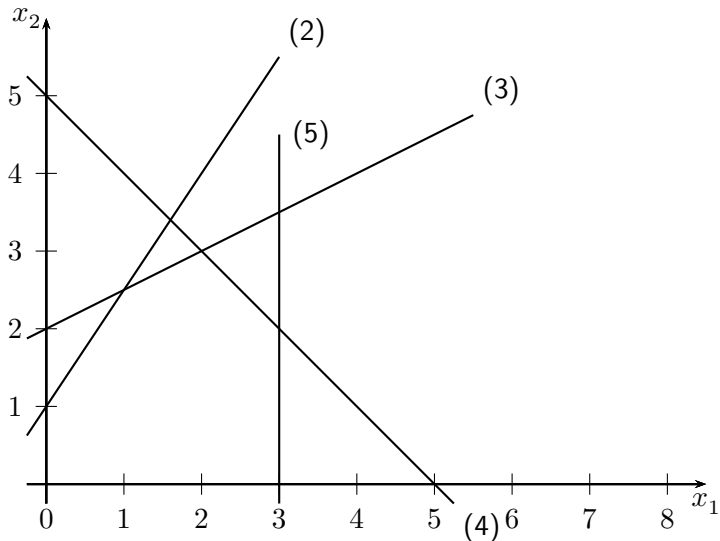
Graphical method



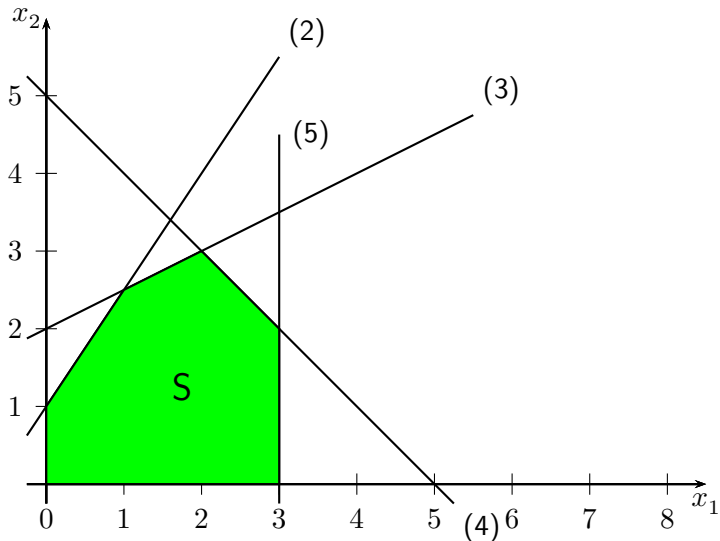
Graphical method



Graphical method



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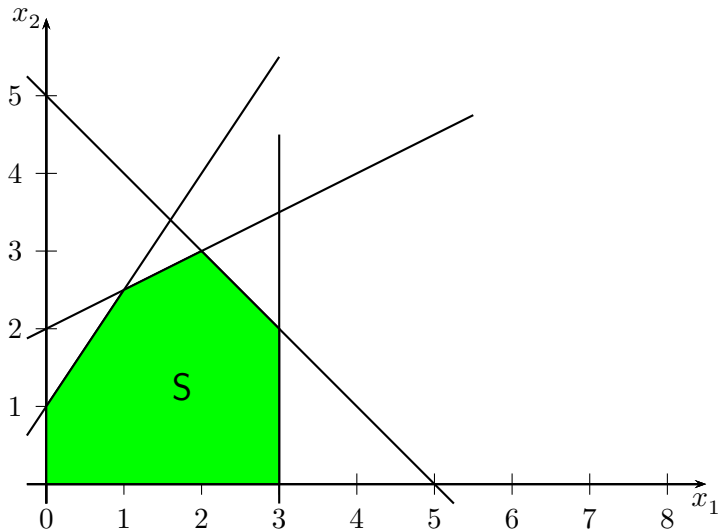
Then, the objective function (1) is plotted with $Z = 0$.

In a **maximization** problem, an optimal solution is found at the intersection of S with a line such that:

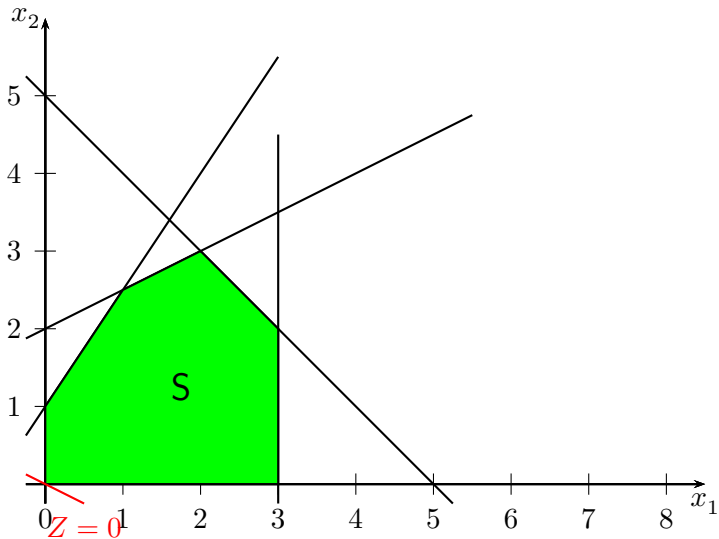
- this line is parallel to the objective function line, and
- this line is the **furthest** one from $(0, 0)$.

The method is the same for a **minimization** problem, but **furthest** is replaced with **closest**.

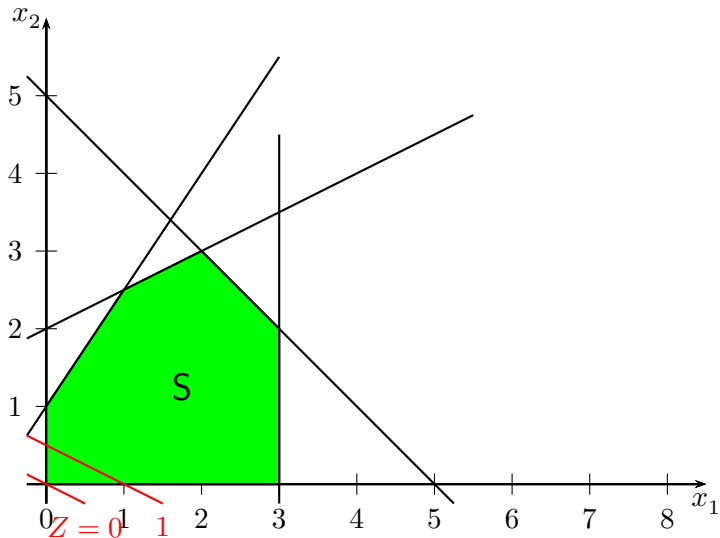
Graphical method



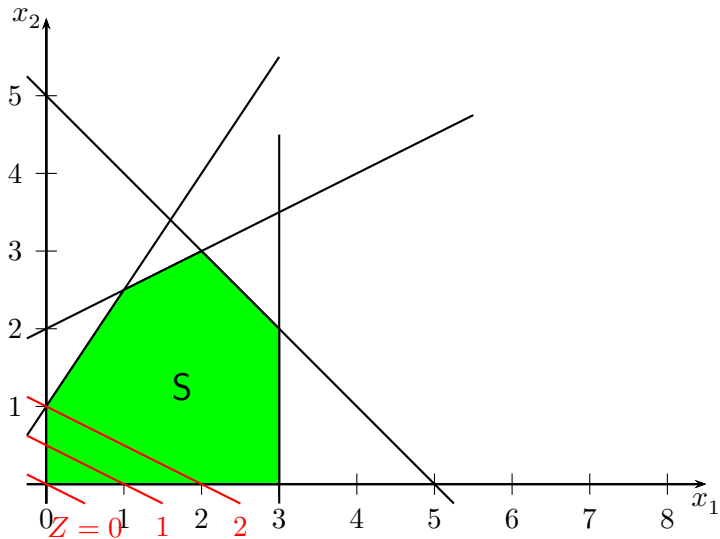
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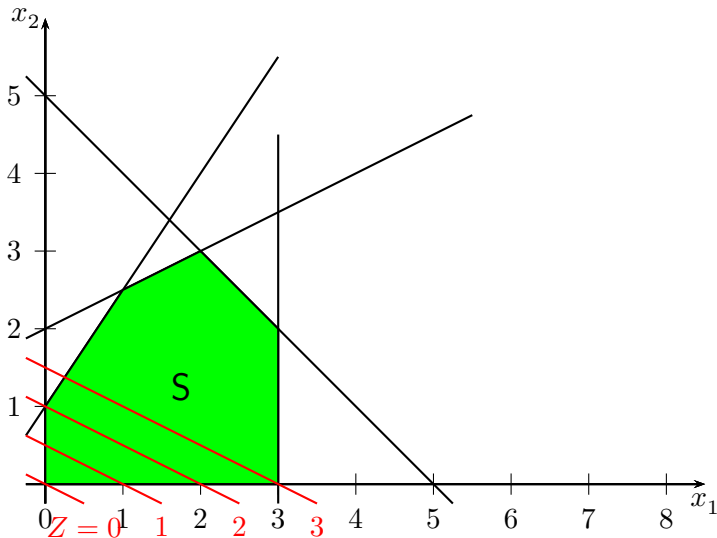
Graphical method



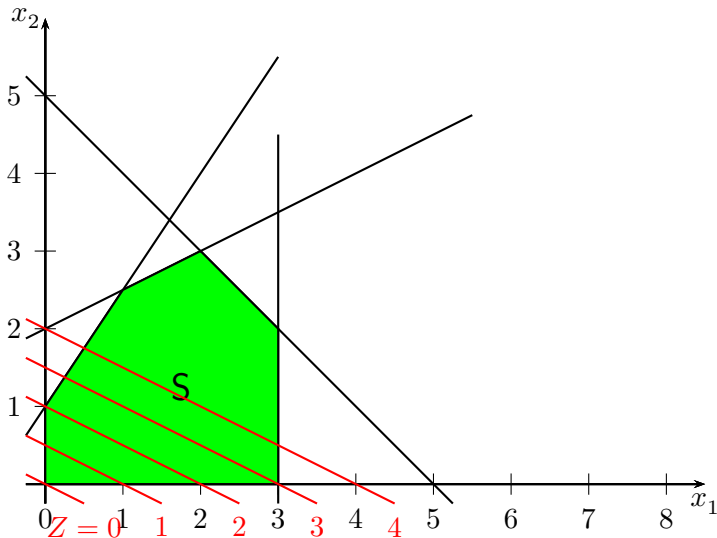
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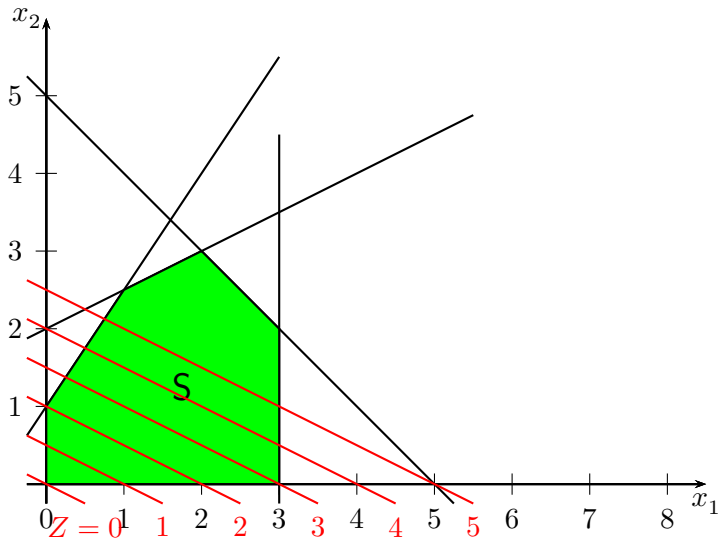
Graphical method



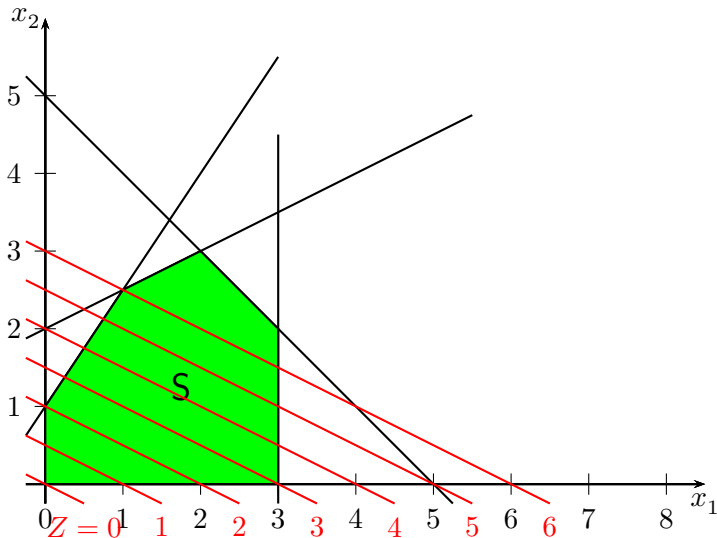
Graphical method



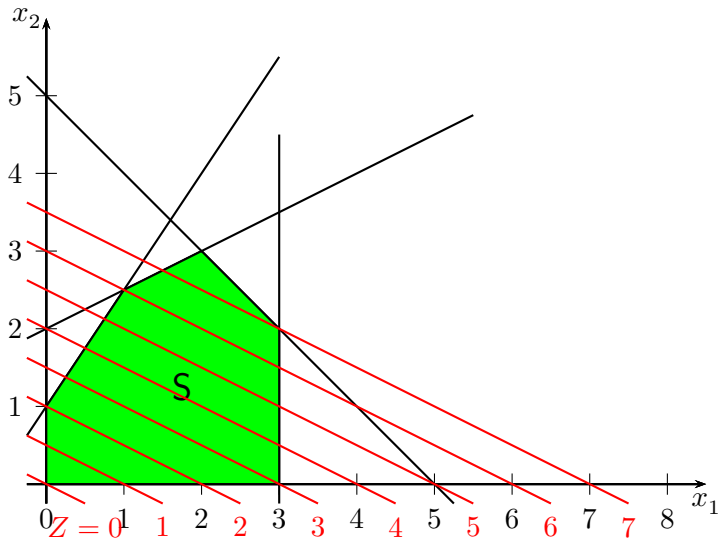
Graphical method



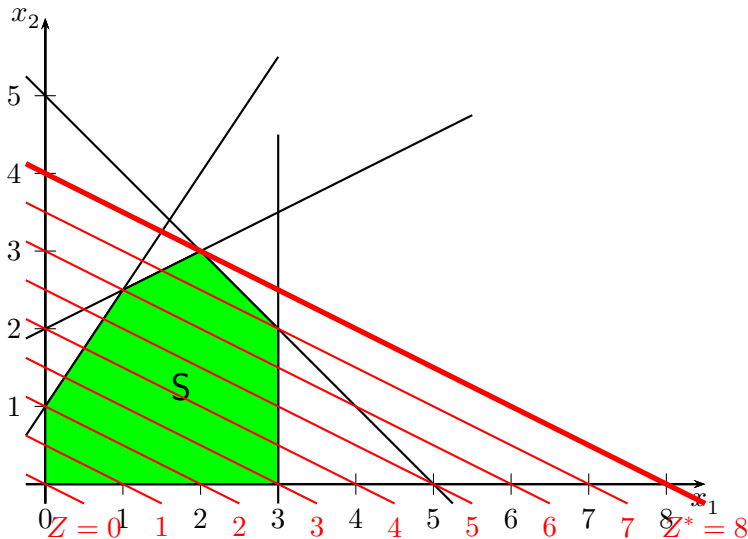
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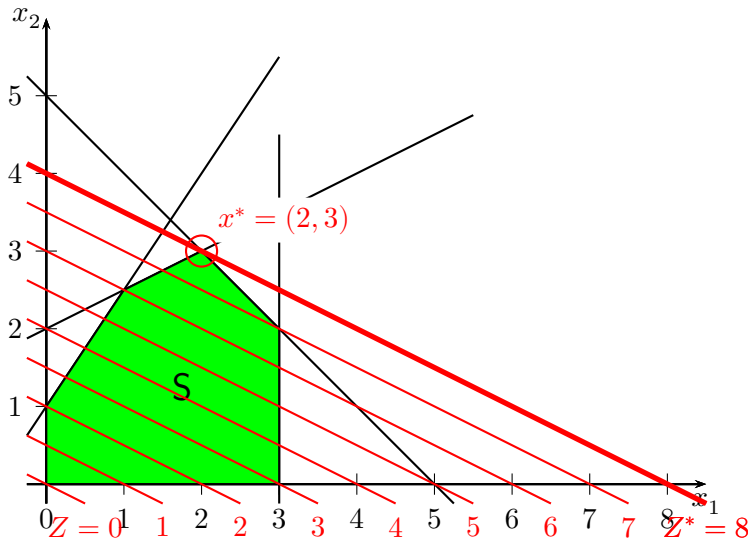
Graphical method



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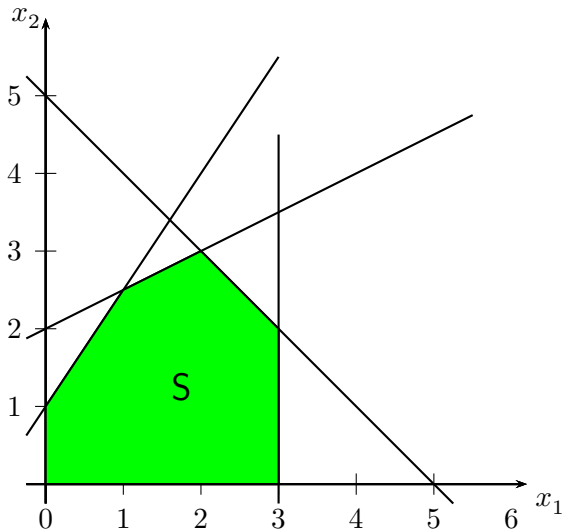
- Move from one vertex to another

Simplex idea

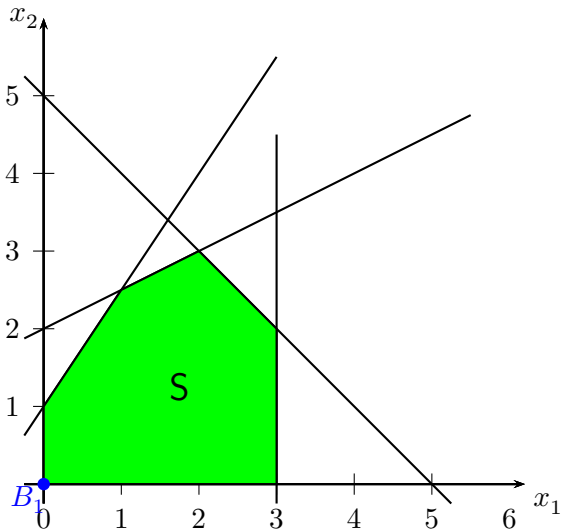
- Move from one vertex to another
- Improve objective step by step

- Move from one vertex to another
- Improve objective step by step
- Stop when no improvement possible

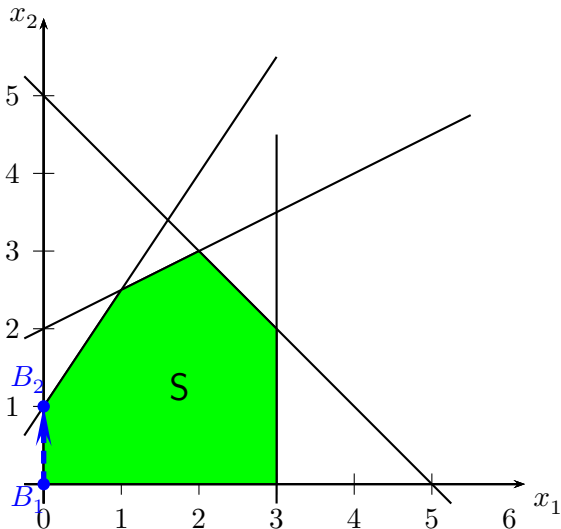
The Simplex on the example



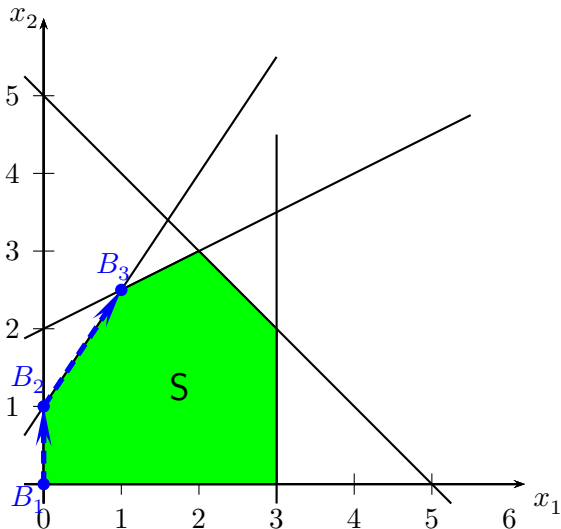
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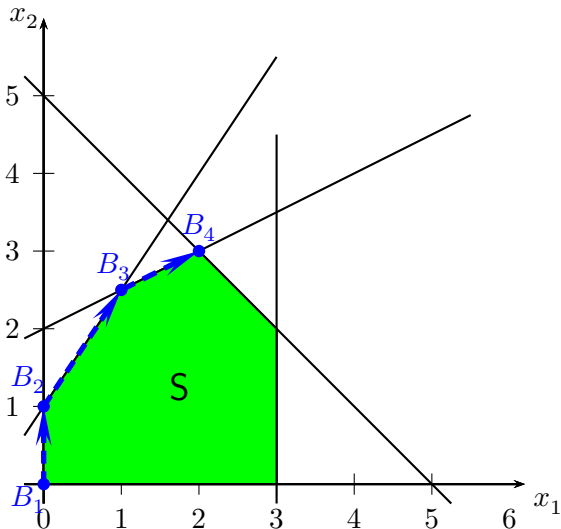
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julia example I

```
# Julia example for formation MathSTIC-B0
```

```
# Maximize  $Z = x_1 + 2x_2$ 
```

```
# s.t.
```

```
#  $-3x_1 + 2x_2 \leq 2$ 
```

```
#  $-x_1 + 2x_2 \leq 4$ 
```

```
#  $x_1 + x_2 \leq 5$ 
```

```
#  $x_1 \leq 3$ 
```

```
#  $x_1, x_2 \geq 0$ 
```

```
#-----
```

```
# Reading Packages
```

```
using JuMP, HiGHS
```

```
# --DECLARATION-----  
# Declare model + optimizer  
m = Model(HiGHS.Optimizer)  
  
# Declare variables  
@variable(m, x1 >= 0)  
@variable(m, x2 ≥ 0) # ≥ can be obtained with \ge + TAB  
  
# Declare constraints  
@constraint(m, -3x1 + 2x2 ≤ 2) # ≤ → \le + TAB  
@constraint(m, -x1+2x2 ≤ 4)  
@constraint(m, x1+x2 ≤ 5)  
@constraint(m, x1 ≤ 3)
```

julia example III

```
# Declare objective and Max/Min
```

```
@objective(m, Max, x1+2x2)
```

```
#-----
```

```
print(m) # Print the model for checking
```

```
#--RESOLUTION-----
```

```
optimize!(m)
```

```
println("Termination status: ",termination_status(m))
```

```
#--SOLUTION-----
```

```
println("Objective value is: ",objective_value(m))
```

```
println("x1 = ", value(x1),"; x2 = ",value(x2))
```

```
#-----
```

Now: you work

- We will use Julia + JuMP + HiGHS

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- We will use Julia + JuMP + HiGHS
- Two problems:
 - Assignment problem
 - Facility location

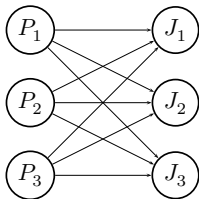
Assignment problem

Suppose we have 3 jobs $\{J_1, J_2, J_3\}$ to be assigned to 3 people $\{P_1, P_2, P_3\}$. The table below models the people fitness for the jobs.

$$c = \begin{array}{ccccc} & J_1 & J_2 & J_3 & \\ \begin{array}{c} P_1 \\ P_2 \\ P_3 \end{array} & \begin{bmatrix} 10 & 4 & 1 \\ 10 & 2 & 7 \\ 8 & 3 & 5 \end{bmatrix} & & & \end{array}$$

$c_{i,j}$ is the fitness of person i for job j . The highest $c_{i,j}$, the more satisfied is person i with job j .

We want to assign a job to each people so as to maximize the total fitness (we seek for the best global fitness).



This can be modeled as a bipartite graph.
Decision variables:

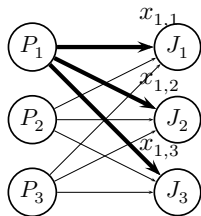
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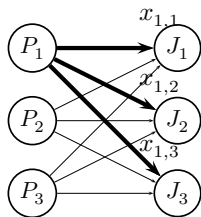
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Decision variables:

$x_{i,j}$ is set to 1 iff person i is assigned job j

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$$\text{Maximize } \sum_{i=1}^n \sum_{j=1}^n c_{i,j} x_{i,j}$$

Assignment problem

The linear program is:

Note: The domain of variables $x_{i,j}$ should be $\{0, 1\}$ (they are Boolean variables) however, these constraints will be met for any optimal solution because of the particular problem structure.

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$$\left\{ \begin{array}{l} \text{Maximize } \sum_{i=1}^n \sum_{j=1}^n c_{i,j} x_{i,j} \\ \sum_{j=1}^n x_{i,j} = 1 \quad \forall i \in \{1, \dots, n\} \\ \sum_{i=1}^n x_{i,j} = 1 \quad \forall j \in \{1, \dots, n\} \\ x_{i,j} \geq 0 \quad \forall i \in \{1, \dots, n\} \forall j \in \{1, \dots, n\} \end{array} \right.$$

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Facility location: example with capacities

A company wants to open distribution centers to serve customers.

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Customers (demand):

Customer	Demand
C_1	5
C_2	7
C_3	4
C_4	6

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C_2	7
C_3	4
C_4	6

Potential facility locations (capacity & opening cost):

Facility	Capacity	Opening cost
F_1	10	30
F_2	15	40
F_3	12	35

Service costs c_{ij} :

	F_1	F_2	F_3
C_1	10	20	15
C_2	12	18	14
C_3	8	25	20
C_4	15	10	12

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Question: Which facilities to open to serve and satisfy customers' demand and the lowest cost?

Think before modeling

Question 1

Can we serve all customers with only one facility?

No. Total demand is:

$$5 + 7 + 4 + 6 = 22$$

The largest facility capacity is 15, so one facility is not enough.

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Question 2

Which facility would you open first?

A natural first choice is F_2 because it has the largest capacity:

$$15$$

and it is attractive for customer C_4 .

Question 3

If you open F_1 , which customers would you assign to it?

Facility F_1 has capacity 10.

Customer	Demand	Cost to F_1
C_1	5	10
C_2	7	12
C_3	4	8
C_4	6	15

A good choice is:

$$C_1 + C_3$$

because:

$$5 + 4 = 9 \leq 10$$

Question 4

Why might a cheap facility NOT be a good choice?

Because it may have limited capacity or high service costs for some customers.

Decision variables

$$y_j = \begin{cases} 1 & \text{if facility } F_j \text{ is opened,} \\ 0 & \text{otherwise,} \end{cases}$$

$$x_{ij} = \begin{cases} 1 & \text{if customer } C_i \text{ is assigned to facility } F_j, \\ 0 & \text{otherwise.} \end{cases}$$

Facility location: constraints

Each customer is assigned to exactly one facility:

$$\sum_{j \in J} x_{ij} = 1 \quad \forall i \in I$$

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$$x_{ij} \in \{0, 1\}, \quad y_j \in \{0, 1\}$$

Objective

$$\min \sum_{j \in J} f_j y_j + \sum_{i \in I} \sum_{j \in J} c_{ij} d_i x_{ij}$$